```
vboxuser@Parul:~$ nano exp7a.c
vboxuser@Parul:~$ gcc exp7a.c -o exp7a
vboxuser@Parul:~$ ./exp7a
CPU task running: 100000000
CPU task running: 200000000
CPU task running: 300000000
CPU task running: 400000000
CPU task running: 500000000
CPU task running: 600000000
CPU task running: 700000000
CPU task running: 800000000
CPU task running: 900000000
```

vboxuser@Parul: ~ — nano exp7a.c

```
  GNU nano 8.7.1
#include<stdio.h>

int main(){
long long counter = 0;
while(1){
counter++;
if(counter % 100000000 == 0){
printf("CPU task running: %lld\n", counter);
}
}
return 0;
}
```

vboxuser@Parul: ~ — nano exp7b.c

```
  GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
int main(){
FILE *fp;
printf("Starting I/O task...\n");
while(1){
fp = fopen("dummy.txt", "w");
if(fp!= NULL){
fprintf(fp, "Writing disk data\n");
fclose(fp);
printf("File operation done\n");
}
sleep(1);
}
return 0;
}
```

```
vboxuser@Parul:~$ pgrep -a exp7a
10988 ./exp7a
vboxuser@Parul:~$ pmap -x 10988
10988:   ./exp7a
Address           Kbytes     RSS   Dirty Mode  Mapping
0000565935e5e000       4       4       0 r---- exp7a
0000565935e5f000       4       4       0 r-x-- exp7a
0000565935e60000       4       4       0 r---- exp7a
0000565935e61000       4       4       4 r---- exp7a
0000565935e62000       4       4       4 rw--- exp7a
000056595deff000     132       4       4 rw---   [ anon ]
00007794aa800000     160     160       0 r---- libc.so.6
00007794aa828000    1632    1056       0 r-x-- libc.so.6
00007794aa9c0000     312     184       0 r---- libc.so.6
00007794aaa0e000      16      16      16 r---- libc.so.6
00007794aaa12000       8       8       8 rw--- libc.so.6
00007794aaa14000      52      20      20 rw---   [ anon ]
00007794aabb5000      12       8       8 rw---   [ anon ]
00007794aabc8000       8       4       4 rw---   [ anon ]
00007794aabca000      16       0       0 r----   [ anon ]
00007794aabce000       8       0       0 r----   [ anon ]
00007794aabd0000       8       8       0 r-x--   [ anon ]
00007794aabd2000       4       4       0 r---- ld-linux-x86-64.so.2
00007794aabd3000     188     188       0 r-x-- ld-linux-x86-64.so.2
00007794aac02000      44      44       0 r---- ld-linux-x86-64.so.2
00007794aac0d000       8       8       8 r---- ld-linux-x86-64.so.2
00007794aac0f000       4       4       4 rw--- ld-linux-x86-64.so.2
00007794aac10000       4       4       4 rw---   [ anon ]
00007ffc7704d000     132      12      12 rw---   [ stack ]
ffffffffff600000       4       0       0 --x--   [ anon ]
---------------- ------- ------- -------
total kB            2772    1752      96
```

```
Enter numbers of students:2
Enter roll number:101
Enter student name:Dhriti
Enter marks:35
Enter roll number:105
Enter student name:Meenakshi
Enter marks:45
Student Database:
(101, 'Dhriti', 35.0)
(105, 'Meenakshi', 45.0)
PS C:\Users\Parul\OneDrive\Desktop\Python>
```

```
Enter number of students:2
Enter roll number:101
Enter student name: Radhika
Enter marks:44
Enter roll number:104
Enter student name: Manish
Enter marks:42
Enter roll number to serach:104
Name: Manish
Marks: 42.0
PS C:\Users\Parul\OneDrive\Desktop\Python>
```

```
vboxuser@Parul:~$ pgrep -a exp7b
11053 ./exp7b
vboxuser@Parul:~$ top -p 11053

top - 07:16:12 up  5:22,  1 user,  load average: 0.69, 0.39, 0.23
Tasks:   1 total,   0 running,   1 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.5 us,  0.7 sy,  0.1 ni, 98.6 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem :   2200.2 total,    258.6 free,   1351.5 used,    799.9 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    848.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  11053 vboxuser  20   0    2768   1692   1592 S   0.0   0.1   0:00.04 exp7b
```

```
vboxuser@Parul:~$ pgrep -a exp7a
vboxuser@Parul:~$ pgrep -a exp7a
11013 ./exp7a
vboxuser@Parul:~$ top -p 11013

top - 07:12:01 up  5:18,  1 user,  load average: 0.67, 0.31, 0.18
Tasks:   0 total,   0 running,   0 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.1 us,  0.2 sy,  0.0 ni, 99.7 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   2200.2 total,    255.6 free,   1355.5 used,    798.9 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    844.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
```